Ogbonna Arinze

arinzeogbo@gmail.com | LinkedIn

SUMMARY

Senior AI-Native Fullstack Engineer with 8 years of professional experience building scalable web applications and integrating generative AI into production systems. Expert in modern frontend architecture (React, Next.js, TypeScript) and robust backend development (NodeJS, NestJS, REST APIs), with a strong foundation in PHP from early career projects. Proven track record of designing distributed systems, containerized deployments (Docker, CI/CD pipelines), and multi-agent LLM pipelines (OpenAI, LangChain, RAG). Adept at driving end-to-end product ownership, evaluating new AI tools, and collaborating across cross-functional teams to deliver high-performance SaaS solutions.

CORE COMPETENCIES

- **AI & LLM Integration:** OpenAI API, LangChain, Vercel AI SDK, Mastra, RAG pipelines, vector search (pgvector, Chroma), agentic workflows, and prompt engineering.
- **Frontend Architecture:** React, Next.js, TypeScript, state management (Redux, Zustand, React Query), component architecture, and Tailwind CSS.
- **Backend & Distributed Systems:** NodeJS, NestJS, REST API design, microservices, Nx monorepos, PostgreSQL, MongoDB, and Prisma.
- **Programming Languages:** TypeScript, JavaScript (ESNext), NodeJS, Python, and PHP.
- **Cloud & Containerization:** Docker, Docker Compose, GitHub Actions, CI/CD pipelines, AWS, Vercel, and Heroku.
- **Testing & Quality:** Playwright, Jest, Cypress, ESLint, Prettier, and CI-driven QA.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating a legacy application to a **React** Chrome extension, optimizing workflows for clinical site integrations and enhancing adoption.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling real-time conversational data querying.
- Implemented **JWT-based authentication** and role-based access control in Next.js with **TypeScript** and React Query, ensuring secure token handling and permission-based UI rendering.
- Optimized application performance by implementing lazy loading, code splitting, and efficient asset delivery, significantly reducing load times as measured via Lighthouse.
- Increased release reliability by integrating **Playwright** end-to-end tests into the **CI/CD pipeline**, automating reporting with Allure Reports on GitHub Actions.
- Architected a reusable design system in React and Radix UI that improved UI consistency, accessibility, and developer efficiency across products.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a 25% boost in user engagement and collaborating with architects to create a scalable frontend architecture.
- Optimized frontend performance by profiling render bottlenecks with React Profiler and Lighthouse, reducing bundle size by 30-40% and achieving a 35% faster load time.
- Drove a 40% improvement in application performance by optimizing caching mechanisms across the React application.
- Established engineering standards using ESLint, Prettier, Husky, Lint-staged, and **GitHub Actions**, reducing errors and accelerating **CI/CD** processes.
- Enhanced development workflow by introducing Jest and Enzyme testing frameworks, leading to a 20% decrease in bug reports.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across cross-functional teams.

Founders Lair — Beijing, China (Remote)
Full Stack Engineer (Contract) — *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** to power new features, focusing on testable, maintainable, and well-documented code.
- Translated business requirements into efficient MongoDB database models and schemas, collaborating closely with the business team.
- Transformed user journeys and Figma designs into reusable **React** components, ensuring a consistent, intuitive user interface.

- Engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on Heroku and production servers on AWS.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed MongoDB database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed SureEdu, a school management portal in a team of 5, handling staff payroll, student records, and guardian communications.
- Responsible for full-stack integration, connecting the JavaScript frontend to NodeJS backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected a distributed system as an **Nx monorepo** managing 6 applications: **NestJS** API backend, **Next.js** citizen-facing web app, admin dashboard, document ingestion pipeline, marketing site, and a video generation service.
- Designed and implemented a **multi-agent AI system** using the Mastra framework with 5 specialized agents for intent classification and domain-specific **RAG-powered analysis** using **OpenAI API**.
- Built a hybrid RAG pipeline combining **pgvector** similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion, with **Cohere** reranking and Voyage AI embeddings.
- Configured **CI/CD** with **GitHub Actions** for automated builds and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and graceful shutdown.
- Implemented SSE streaming for real-time chat responses, PostgreSQL 16 database schema with Prisma v7, and custom migration tooling.
- Built the Next.js frontend with **React 19**, Tailwind CSS, and Recharts, featuring 22 interactive chart types auto-generated from AI responses.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science, 4.1 CGPA *Graduation Date: Sep 2019*